

## NOVELTY OF URBANFIX

UrbanFix does not rely on a single isolated technology such as only AI image classification, only NLP complaint classification, or only a geo-based complaint reporting system.

Its novelty lies in integrating the major capabilities proposed independently across multiple civic complaint management approaches into one unified, end-to-end smart governance ecosystem.

## KEY NOVEL CONTRIBUTIONS

### 1. MULTIMODAL COMPLAINT REGISTRATION

- Text-based complaint registration
- Image-based complaint registration
- AI-assisted complaint understanding

### 2. AI + QUESTIONNAIRE-BASED PRIORITY SYSTEM

- AI image analysis
- Detected issue type
- Visual severity
- Dynamic citizen questionnaire
- Location-based importance
- Duplicate complaint frequency

These signals are combined instead of assigning priority using only one parameter.

### 3. HYBRID DUPLICATE DETECTION

- Textual similarity
- Image similarity
- Geographic proximity
- Temporal proximity

This provides a stronger duplicate detection mechanism than using only cosine similarity on complaint text.

### 4. CLOSED-LOOP CIVIC ISSUE LIFECYCLE

DETECTION



AI CLASSIFICATION



PRIORITY CALCULATION



DUPLICATE DETECTION



DEPARTMENT ROUTING



WORKER ASSIGNMENT



GPS NAVIGATION



REAL-TIME PROGRESS



BEFORE/AFTER EVIDENCE



PUBLIC VERIFICATION



PREDICTIVE ANALYTICS

## 5. TWO TYPES OF ANALYTICS

- Temporal activity heatmaps
- Geographic complaint-density heatmaps

## 6. HUMAN + AI COLLABORATION

- AI suggests the category
- AI estimates severity
- AI identifies potential duplicates
- AI recommends priority and department
- Citizens and administrators retain control for verification and correction

## 7. TRANSPARENT AND AUDITABLE AI-GOVERNED WORKFLOW

- Existing immutable status history
- Mandatory administrative remarks
- Public complaint registry
- PII redaction
- Resolution receipts
- AI decision metadata

## 8. PREDICTIVE + REACTIVE GOVERNANCE

#### REACTIVE:

Citizens report existing problems.

#### PROACTIVE:

Historical complaint data is analyzed to predict recurring issues and identify geographic hotspots before they become large-scale problems.

### CORE NOVELTY STATEMENT

"UrbanFix proposes a unified AI-driven, geospatially aware and publicly transparent civic complaint management ecosystem that combines computer vision, NLP, hybrid duplicate detection, multi-factor priority assessment, intelligent departmental routing, worker navigation, predictive analytics, and verifiable before-and-after resolution evidence within a single end-to-end governance workflow."

### CURRENT TECH STACK

#### FRONTEND

- React 19
- Vite
- React Router v6
- Axios
- Chart.js v4
- react-chartjs-2 v5
- Lucide-react
- Geist Fonts
- Custom CSS / Hand-rolled CSS Design System

#### BACKEND

- Node.js
- Express.js
- ES Modules
- Morgan

#### DATABASE

- MongoDB
- Mongoose v8

#### AUTHENTICATION & SECURITY

- JSON Web Tokens (JWT)
- bcryptjs
- Email OTP Verification

- Zod Validation

#### FILE HANDLING

- Multer
- Multipart Form Data
- Image Upload Support

#### EMAIL

- Nodemailer
- SMTP
- Ethereal Test SMTP Fallback

#### DOCUMENT GENERATION

- PDFKit

#### CURRENT ARCHITECTURE

- React + Vite Frontend
- REST API Communication
- Express.js Backend
- MongoDB Database
- Mongoose ODM